Computational Analysis of Differential Gene Expression Within Alzheimer's Disease and Healthy Brain Samples

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide. It's known to deteriorate memory, language, and even physical movement. The accumulation of amyloid-beta plaques and neurofibrillary tau tangles in the brain is what's believed to cause the disease (Abdulkhaliq et al., 2026). This accumulation disrupts neuronal communication and leads to cell death. Environmental and lifestyle factors also play a role in disease risk, but genes seem to be the main predictor. The strongest known genetic risk factor is the APOE ε4 allele, and is believed to influence amyloid clearance (Hunsberger et al., 2019). Understanding which genes are differentially expressed between healthy and AD brain tissue is important because it will ultimately help point to the biological pathways that drive the disease. Here, I analyze a publicly available RNA-sequencing dataset, accession number GSE104704, from the Gene Expression Omnibus (GEO). This dataset contains samples from the lateral temporal lobe of eight healthy young brains, ten aged brains, and twelve AD brains. For the sake of this project, I combined both young and old healthy brains into one group for analysis.

I used three computational approaches: volcano plots, principal component analysis (PCA), and Uniform Manifold Approximation and Projection (UMAP) combined with K-means clustering. Throughout the methods, data was verified by printing out the shapes of the data, depending on the plot, and comparing results with the GEO analysis plots. The volcano plot was used to identify genes that are statistically differentially expressed between AD and healthy samples. The PCA was used to determine if there were any batch effects between the samples. Lastly, UMAP and K-means clustering were used to determine whether AD and healthy samples formed distinct clusters based on their different overall genetic expressions. I hypothesize that two clusters will form, where one will be for the healthy samples and the other for AD samples. Additionally, I hypothesize this will be due to genetic differences in disease status rather than age.

The volcano plot identified 2,870 statistically significant differentially expressed genes (Figure 1A). The most upregulated genes in AD include LINC02937 (100507103), TOP3B (8940), TEKT2 (27285), LYG1 (129530), and LOC105369968 (105369968). Most notably, TOP3B is responsible for the regulation of the topological state of DNA during transcription, replication, and repair by temporarily breaking and rejoining a single DNA strand and suggests that AD brain tissue could be experiencing more transcriptional stress or altered gene regulation. This is consistent with the known dysregulation of gene expression networks in neurodegeneration (Joo et al., 2020). Moreover, TEKT2 is part of the tektin family, which is a structural component of ciliary and flagellar microtubules. Lastly, LYG1 is a lysozyme-related gene that is linked to immune response. The downregulated genes expressed in the volcano plot include RABEPK (10244), LOC10795001 (10795001), DPH2 (1802), GLUD2 (2747), TXN2 (25828). Most notably, GLUD2 is involved in glutamate metabolism. Dysregulation of glutamate signaling is known in AD and is associated with excitotoxicity (Wood et al., 2022). Additionally, RABEPK is essential in late endocytotic trafficking and vesicle transport between organelles. Impaired vesicle transport has been linked to defective amyloid precursor protein processing and increased amyloid-beta production. TXN2 influences cell signaling, adhesion, migration, and differentiation, which are all important in maintaining neuronal connectivity and synaptic integrity.

PCA was used to determine if there were batch effects within the data. PC1 explained 25.2% and PC2 explained 20.5% of variance, for a total of 45.6% explained variance (Figure 1C). Additionally, the samples didn't form clear separations by disease state, they were much more mixed together (Figure 1B). This suggests that the data isn't necessarily driven by disease state and there was also no batch effect between the two groups.

UMAP was used to visualize gene expression in a two dimensional space. This is a better method at preserving local structure and interpreting nonlinear data. The results demonstrate partial separation between the AD and healthy groups (Figure 2A). AD tended to group more towards the left bottom corner of the plot, while the healthy samples clustered more towards the middle and right regions of the plot. It's important to note that despite these differences, the groups did not form distinct clusters. K-means clustering with k=2 was applied to the UMAP to quantitatively assess clustering. The silhouette score was 0.387 which points to a more moderate, but not strong cluster separation (Figure 2B). Additionally, the K-means clusters agreed with the true labels, consistent with most of the overlap observed in the UMAP. These results suggest that the genetic differences between the AD and healthy samples are detectable, but not different enough to separate into distinct clusters at the sample level.

This project used volcano plots, PCA, and UMAP with K-means clustering to compare gene expression between AD and healthy lateral temporal lobe brain tissue. The volcano plot was able to identify 2,870 significantl differentially expressed genes. These genes are important factors in many different biological processes like: neuroiflammation, glutamate metabolism, vesicle trafficking, and oxidative stress. PCA confirmed there was no batch effect between the samples. The UMAP and K-means silhouette score of 0.387 suggest there is no clear clustering between the samples and disproves the initial hypothesis. This could be due to unique individual gene expression profile regardless of disease state. Bulk RNA sequencing also has limitations as it sequences all cel types together, which could diminish the signals sent from neurons. Additionally, the dataset only had a total of 30 samples, which makes it difficult to clearly separate the samples. Future research could instead focus on investigating single-cell RNA sequencing so that those genetic differences aren't overshadowed. This research holds the potential to contribute biomarkers or therapeutic targets for AD, which currently has no cure.

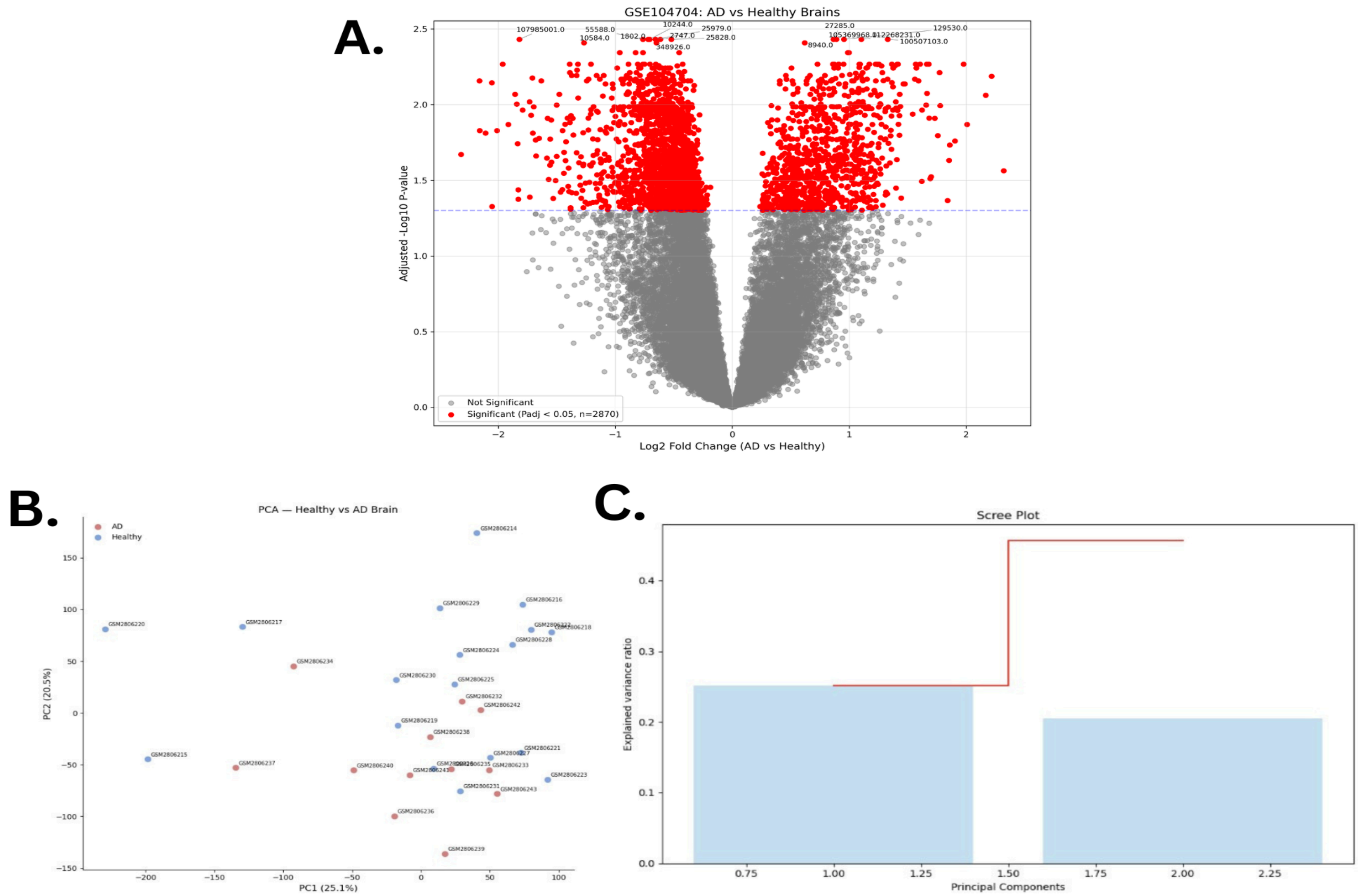
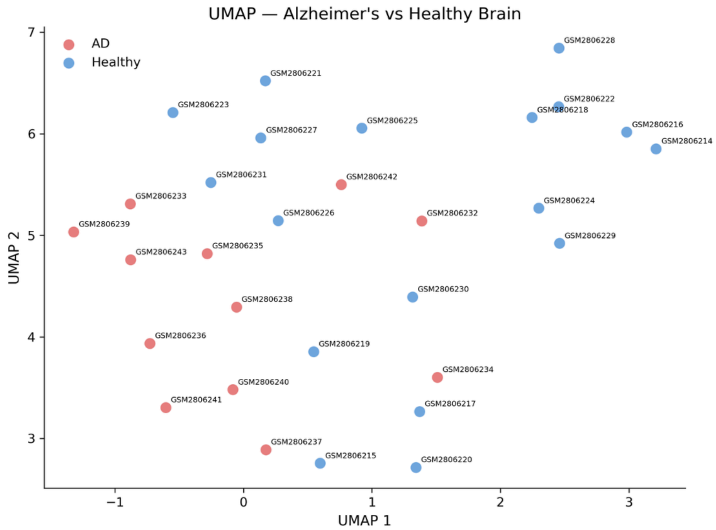


Figure 1. Differential gene expression and dimensionality reduction analysis of GSE104704. **(A)** Volcano plot comparing Alzeimers Disease (AD) with healthy brain samples. Each point represents one gene. Red points are statisticaly significant. The y-axis demonstrates the -log10 adjusted p-value and the x-axis represents the log2 fold change between mean expression. **(B)** Principal component analysis (PCA) of all 30 samples. PC1 explains 25.1% and PC2 explains 20.5% of total variance. The red dots represent AD samples, while blue represents healthy samples. **(C)** Scree plot showing the explained variance ratio for each principal component. The cumulative variance is shown in red.

A.



B.

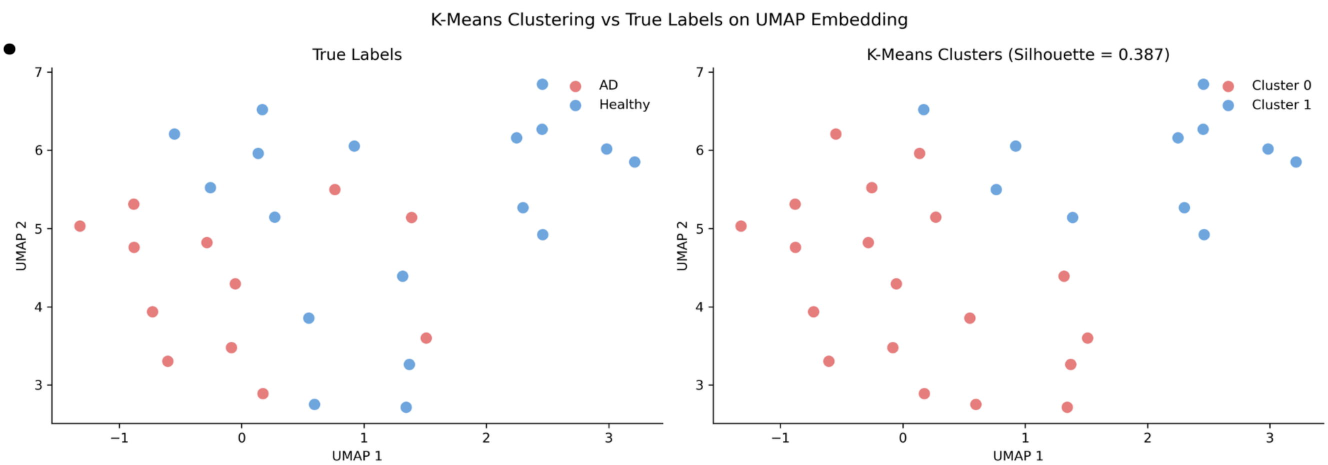


Figure 2. UMAP dimensionality reduction of gene expression data from GSE104704. (A) UMAP embedding of 30 brain samples (18 healthy and 12 AD). Each point represents one sample that is labeled by its GEO accession ID. AD samples are shown in red, while healthy samples are shown in blue. **(B)** Comparison of true labels versus the K-Means clustering on the same UMAP. The silhouette score was 0.387.

References

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AI Statement

Note: AI was used in this project to research the top 15 upregulated and downregulated genes found in the volcano plot and their role within AD.